

Name:

Teacher:



Year 9 Mathematics (5.2) Yearly Examination 2023

Instructions

- Working time: **50 minutes**
- Attempt all questions
- Write in black or blue pen, draw diagrams, tables and graphs in pencil
- NESA approved calculators are allowed
- Show relevant working out where necessary

Surface Area And Volume	Trigonometry	Probability	Total
/15	/15	/15	/45

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Show working where necessary.

Surface Area And Volume

1. Convert:

3

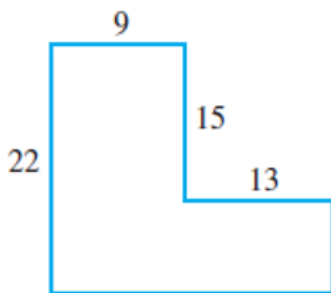
a) 9 cm to mm: _____

b) 2790 g to kg: _____

c) 0.73 L to mL: _____

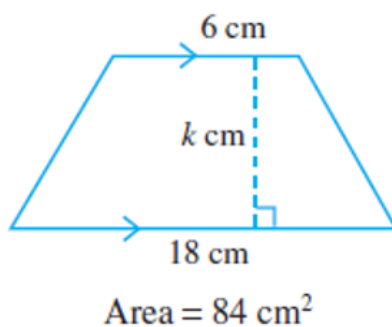
2. Calculate the perimeter of the composite shape. All measurements are in centimetres.

1



3. Find the value of the pronumeral, k .

1

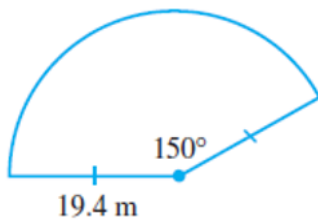


Section 1 continues on the following page

Show working where necessary.

4. For the sector below, find:

2

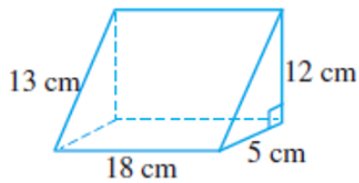


a) The perimeter, correct to two decimal places.

b) The area, correct to two decimal places.

5. Find the surface area of the triangular prism.

2



6. Find the width of a rectangular prism whose length is 13 cm , height is 7 cm and surface area is 542 cm^2 .

2

Show working where necessary.

7. Find the capacity of a saucepan with a diameter of 19 cm and a height of 8 cm . Answer in litres, correct to one decimal place. **2**

8. The paper label on a can of apricot nectar is 26.7 cm long and 8.7 cm high. It completely covers the curved surface without overlap.

a) Find the radius of the can, correct to 1 decimal place. **1**

b) Find the total surface area of the can, correct to the nearest square centimetre. **1**

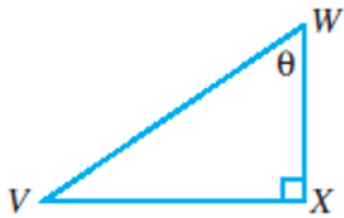
End of Section 1

Show working where necessary.

Trigonometry

1. For the triangle below, with respect to θ , name:

3



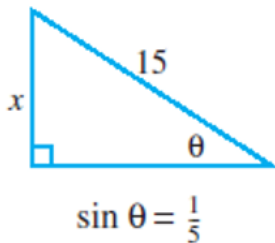
a) the hypotenuse: _____

b) the opposite side: _____

c) the adjacent side: _____

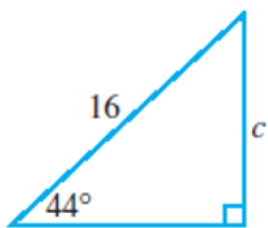
2. Find the value of the pronumeral, x .

1



3. Find the value of c , correct to two decimal places.

1



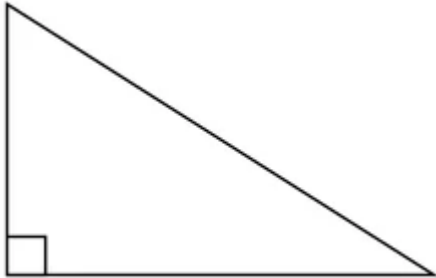
Section 2 continues on the following page

Show working where necessary.

4. Mark all the given information below on the right-angled triangle.

a) In $\triangle LMN$, $\angle M = 90^\circ$, $\angle L = 16^\circ$ and $LM = 34\text{ m}$.

1



b) Find LN , correct to the nearest metre.

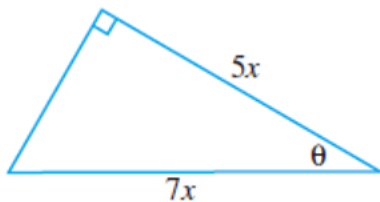
1

5. At a local shopping centre, a wheelchair ramp is inclined to the horizontal at an angle of 14° . Find the length of the ramp if it leads to an entrance of 2.4 m higher than the surrounding floor area. Answer correct to one decimal place.

2

6. Find θ , correct to the nearest degree.

1

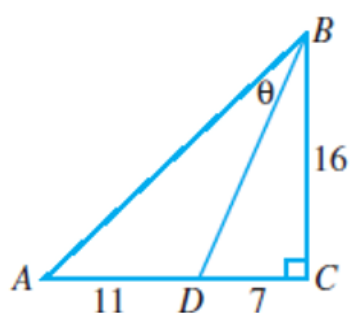


Section 2 continues on the following page

Show working where necessary.

7. A step ladder has legs of length 2.2 m and the maximum distance between the legs is 0.9 m . Find the largest possible angle in which the ladder can be opened. 2

8. Find θ , correct to the nearest degree. 3



End of Section 2

Show working where necessary.

Probability

1. A letter is chosen from the word SQUIDWARD. Find the probability of choosing the letter "D". **1**

2. From the 48 people surveyed at a gaming convention, 36 like Fortnite, 16 like Roblox and 6 like Fortnite and Roblox.

a) Represent the information given in the two-way table. **2**

	Likes Roblox	Does Not Like Roblox	Total
Likes Fortnite			
Does Not Like Fortnite			
Total			

b) A person from the gaming convention is chosen at random. Find the probability that the person: **2**

i) Likes only Roblox?

ii) Likes Roblox or Fortnite, but not both?

Section 3 continues on the following page

Show working where necessary.

3. A drawer contains 2 red socks (R), 1 blue sock (B) and 1 yellow sock (Y). Two socks are selected at random without replacement.

a) Construct a tree diagram to represent the given information, listing all possible outcomes. **2**

b) Find the probability, as a simplified fraction, of obtaining: **3**

i) A red sock and a blue sock

ii) Any pair of socks of the same colour

iii) Any pair of socks of different colour

Section 3 continues on the following page

Show working where necessary.

4. The colour of cars driving along a highway was noted over a short period of time and summarized in the frequency table below.

Colour	White	Silver	Blue	Green
Frequency	7	4	5	4

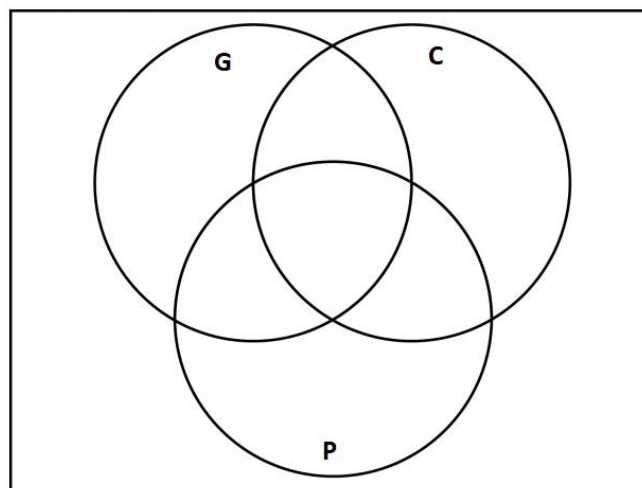
a) Calculate, as a percentage, the experimental probability, of selecting a silver or green car? **1**

b) If the colour of 200 cars was recorded, find the expected number of white cars. **1**

5. A leisure centre surveyed 50 members asking whether they exercise at the gym (G), eat at the cafe (C) or swim in the pool (P). The following points were gathered. (5)

- 29 members used at least one of those facilities
- 15 members eat at the cafe
- 15 members swim in the pool
- 18 members exercise at the gym
- 5 members both used the pool and cafe
- 9 members both used the cafe and gym
- 8 members both used the pool and gym
- 3 members used the gym, cafe and pool

a) Represent the information given in the Venn diagram provided. **2**



b) If a person is chosen at random from the group, what is the probability that the person uses the pool and cafe only? **1**

End of Exam :)

Use this page for working out.